

Master of Data Analytics, University of Niagara Falls, Canada

UNFC Competition

Gabriel Romero Gutierrez - NF 1012849

February 13, 2026

INTRODUCTION

1. Context and Problem Statement

Toronto's waterfront is a vital socioeconomic and recreational asset. However, maintaining safe conditions for public use presents a significant challenge due to the complex interaction between urban infrastructure and environmental variables. This report presents a quantitative framework to prioritize water quality monitoring and public health advisories for Toronto's beaches. By integrating historical *E. coli* measurements with daily weather data (2007–2025), a Multi-Factor Risk Score was developed to categorize sites by their vulnerability to contamination.

METODOLOGY

The reliability of a predictive framework depends on the quality of its inputs. The following multi-stage data cleaning pipeline was implemented to transform raw environmental monitoring data into a robust analytical dataset.

2.1 Data Cleaning and Integrity

To ensure a clean baseline for the analysis, the following steps were executed:

- **Handling Missing Values:** Records with missing *E. coli* concentrations or critical meteorological variables (precipitation/temperature) were removed. Given that water

quality is highly time-sensitive, imputation (filling in gaps) was avoided to prevent the introduction of artificial bias into the bacterial variance.

- **Duplicate Removal:** The dataset was screened for redundant entries (identical site, timestamp, and measurement). No duplicate rows were found during the analysis.
- **Outlier Analysis:** A univariate outlier detection was performed. While high levels of *E. coli* are expected in urban runoff, extreme values can skew the mean and standard deviation, misrepresenting the "typical" risk level of a site. According to the data exploration, *E. coli* levels showed significant skewness. For that reason, a clipping method (IQR-based capping) was applied to prevent extreme outliers from disproportionately biasing the measures while retaining the signal of high-risk events.

2. 2 Analytical Approach & Methodology

The framework followed a four-stage pipeline to ensure data integrity and actionable insights:

Multivariable Structure: The geospatial water quality data were merged with historical weather datasets on a daily temporal grain.

Outlier Management: According to the data exploration, *E. coli* levels showed significant skewness. For that reason, a clipping method (IQR-based capping) was applied to prevent extreme outliers from disproportionately biasing the measures while retaining the signal of high-risk events.

Feature Engineering: An Unsafe Flag was developed, defined as a binary indicator for cases where *E. coli* levels exceed 100 units per 100 ml. This threshold was determined based on the historical standards from the Ontario Recreational Water Quality Guide (PWSO) and Health

Canada recommendations, which is considered the limit to minimize the risk of gastrointestinal illness in swimmers.

Environmental Drivers: The probability of unsafe events under critical conditions was calculated for cases with rainfall greater than 10 mm, based on urban runoff studies that associate this volume with combined sewer overflow (CSO) in the Great Lakes region (Li et al, 2023) Likewise, an average temperature threshold above 22°C was established, based on microbiological literature linking increased water temperature with higher bacterial survival and proliferation rates. (Desta et al, 2024)

The Priority Scoring Model: A weighted **Risk Score (RS)** was designed for each site:

$$RS = (0.5 \cdot \text{Unsafe Rate}) + (0.3 \cdot \text{Norm. Mean } E. coli) + (0.2 \text{ Norm. Sample volume})$$

Where:

- **Unsafe Rate (50%):** Prioritizes frequency of failure (Alert).
- **Mean E. coli (30%):** Accounts for the severity of contamination.
- **Sample Frequency (20%):** Ensures that sites with higher public exposure (more samples) are monitored more strictly.

Priority Classification: Using the RS formula and Quantile Classification, we categorized Toronto's beaches into three priority levels: High, Medium, and Low.

Geospatial Analysis: The data was exported to geospatial analysis software (QGIS) to generate a dynamic alert map. This cartographic representation allows for the spatial visualization of each site's vulnerability, transforming complex metrics into an intuitive visual tool. The ultimate goal

of this mapping is to facilitate more efficient and clear transmission of information, optimizing strategic decision-making and real-time public health alert management.

RESULTS

3.1 Environmental Effect

The analysis shows significant correlations between weather patterns and water safety, highlighting two main drivers. First, precipitation was identified as the most robust predictor of contamination, where under heavy rainfall conditions (>10 mm), the probability of water being classified as unsafe rises to 28.98%, while under light or no rainfall conditions (≤10 mm), this risk drops to 16.23%. This increase in the probability of risk suggests that urban runoff and combined sewer overflows (CSOs) are the critical factors in degradation.

On the other hand, temperature influence showed that days with average temperatures above 22°C have a direct correlation with higher insecurity rates (18.79%), both related to the acceleration of bacterial growth in warm water and to the increase in the number of bathers. Finally, temporal risk shows a clear seasonal peak, with July (19.48%) and August (18.32%) being the most vulnerable months, justifying a dynamic escalation in monitoring efforts during the summer.

Month	Unsafe flag Probability (%)
May	9.76
June	15.75
July	19.49
Agoust	18.32
September	14.66

Table 1. Seasonal unsafe flag probability effect

3.2 Beach Prioritization and Risk Classification

According to the Risk Score (RS) results, beaches were divided into three priority levels. High-Priority areas such as Sunnyside Beach (RS:0.67) and Marie Curtis Park East Beach (RS:0.611)

have the most critical failure rates and the highest sensitivity to rainfall events, requiring daily monitoring and immediate alert protocols. Medium-Priority sites, such as Bluffer's Beach Park, show a moderate risk that is susceptible to being triggered by extreme weather events, justifying routine monitoring. Finally, Low-Priority beaches, such as Gibraltar Point Beach (RS: 0.28), remain consistently safe under most conditions; in these cases, the analytical framework suggests that monitoring efforts can be optimized through less frequent surveillance, allowing for the redistribution of technical and human resources to areas of greater danger to public health.

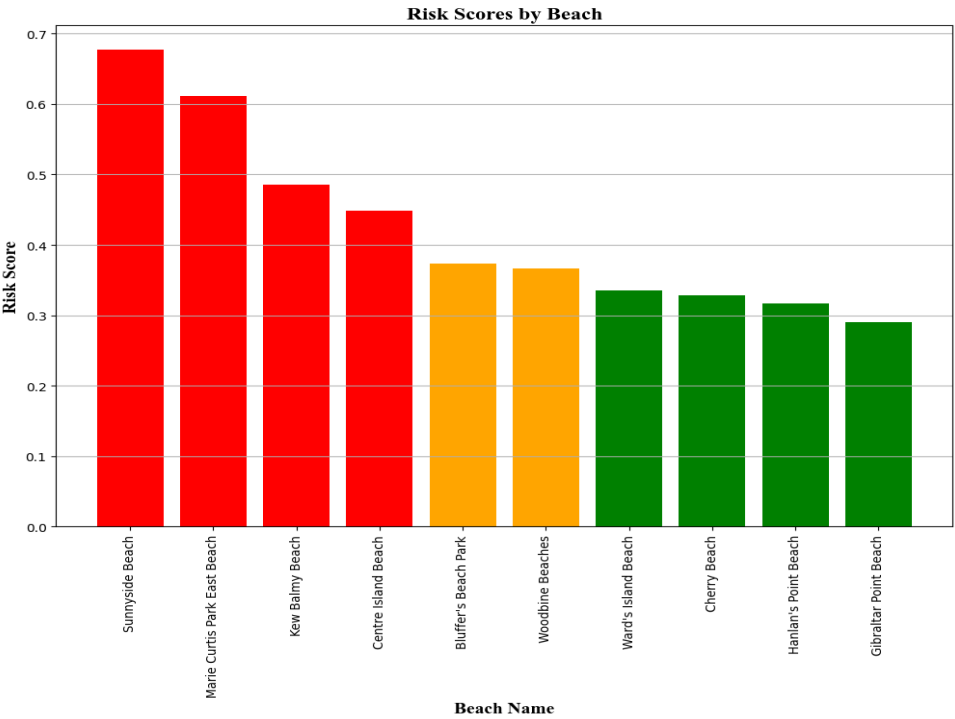


Figure 1. Risk score ranking by beach (RS)

CONCLUSIONS AND RECOMMENDATIONS

The results of this analysis provide a quantitative analysis to support proactive public health management in Toronto. As a primary measure, the implementation of automatic preventive warnings after rainfall events exceeding 10 mm is recommended, given that the probability of

contamination under these conditions significantly rises. It is also imperative to consider prioritizing investments in infrastructure and environmental audits at critical points such as Sunnyside Beach (High Park) and Marie Curtis Park East (Etobicoke), where insecurity rates are the highest, suggesting the presence of persistent sources of contamination.

Beach Name	Location	RS	Priority	Check-in
Sunnyside Beach	High Park	0.67	High	Daily/Critical
Marie Curtis Park East Beach	Etobicoke	0.61	High	Daily/Critical
Kew Balmy Beach	Beaches East-York	0.48	High	Daily/Critical
Centre Island Beach	Toronto Islands	0.44	High	Daily/Critical
Bluffer's Beach Park	Scarborough SW	0.37	Medium	Routine
Woodbine Beaches	Beaches East-York	0.36	Medium	Routine
Ward's Island Beach	Toronto Islands	0.33	Low	Surveillance
Cherry Beach	Toronto Islands	0.32	Low	Surveillance
Hanlan's Point Beach	Toronto Islands	0.31	Low	Surveillance
Gibraltar Point Beach	Toronto Islands	0.28	Low	Surveillance

Table 2 Priority ranking and recommendations

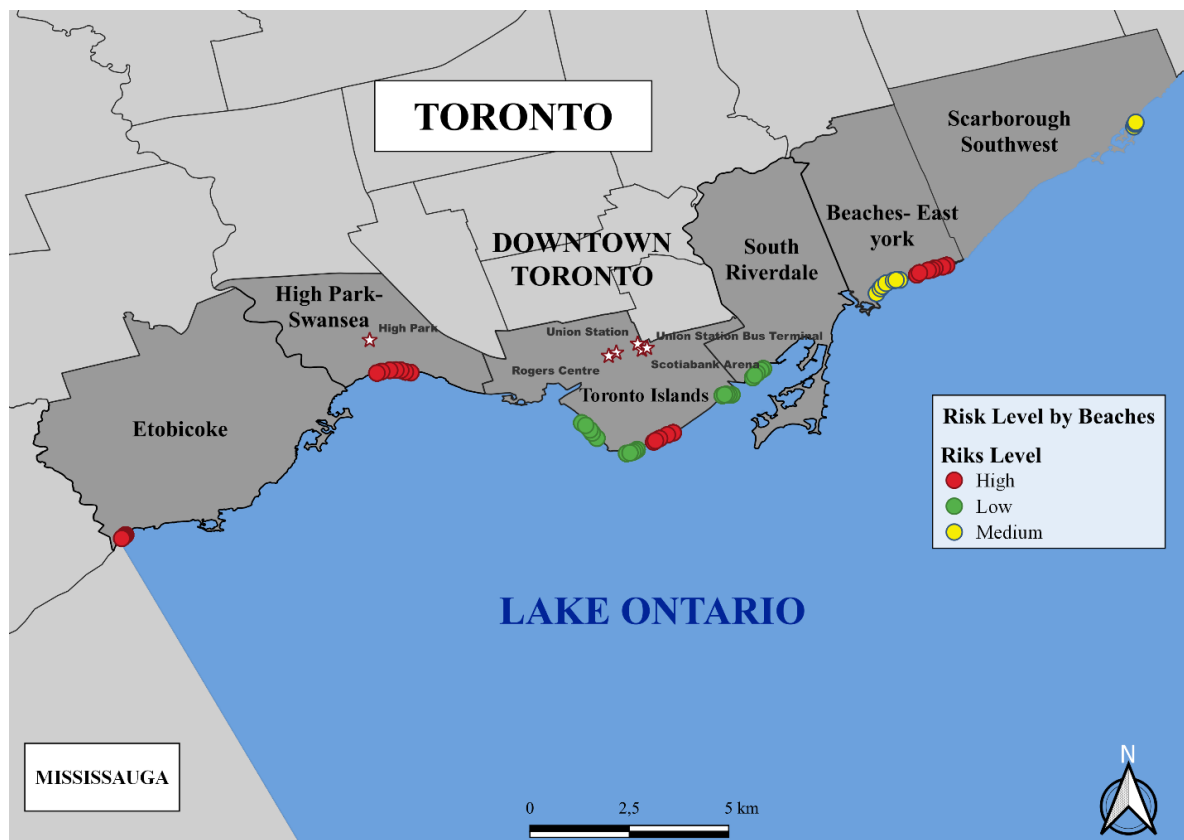


Figure 2. Risk Level beach classification on the Toronto Coastline Area

To optimize operational efficiency, the analytical framework proposes a differentiated scheme: maintaining daily monitoring at “High Priority” sites while transitioning to a biweekly scheme at “Low Priority” beaches such as Gibraltar Point or Ward's Island, reallocating those resources to areas of greater volatility. Finally, public communication strategies should be reinforced during heat waves, a period when water vulnerability coincides with peak beach visitor traffic. The integration of climate and microbiological data into this risk scoring system not only enables more robust and defensible decision-making but also ensures that monitoring efforts are concentrated where the impact on public safety is greatest.

REFERENCES

Herendeen SL, VanBogelen RA, Neidhardt FC.1979.Levels of major proteins of Escherichia coli during growth at different temperatures. J Bacteriol139:.<https://doi.org/10.1128/jb.139.1.185-194.1979>

Rui Li, Gabriel Filippelli, Lixin Wang, Precipitation and discharge changes drive increases in Escherichia coli concentrations in an urban stream, Science of The Total Environment, Volume 886, 2023, 163892, ISSN 0048-9697, <https://doi.org/10.1016/j.scitotenv.2023.163892>.

Shapefiles Regional Municipal Boundary and City Wards were obtained from the Toronto Open Data Source